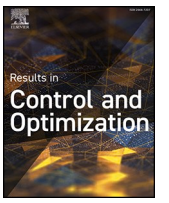
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# Learned prediction of cholesterol and glucose using ARIMA and LSTM models – A comparison

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ABSTRACT

Painless remedies and diagnosis have become the primary objective in medical sciences as diseases shoot up. This research addresses the pressing need for cost-effective, non-invasive, and real-time monitoring solutions for glucose and cholesterol levels to reduce mortality rates associated with diabetes and cardiovascular diseases. Diabetes and cardiovascular diseases are significant global health challenges, affecting millions of individuals, particularly in low to middle-income countries. The inconvenience and discomfort of traditional blood tests have often resulted in delayed testing and monitoring, making it imperative to develop alternative methods for early disease detection and continuous monitoring. The primary objective of this research is to create a non-invasive glucose and cholesterol monitoring and forecasting system that leverages Artificial Intelligence and Cloud computing. The specific focus is comparing the prediction efficiency of two models: the Auto-Regressive Integrated Moving Average (ARIMA) model and the Long Short-Term Memory (LSTM) model. The research methodology involves the development of a hardware prototype for the non-invasive measurement of glucose and cholesterol, employing Near Infra-Red (NIR) sensors. Data on glucose and cholesterol levels are collected and stored from patients over a period of one month. The collected data is used to train both the ARIMA and LSTM models. The models are evaluated using Root Mean Square Error (RMSE), and their prediction accuracy is compared. The study findings reveal that the ARIMA model surpasses the LSTM model in predicting glucose and cholesterol levels. This is supported by a lower RMSE of the ARIMA model (about 71.7 % less for Glucose and 50.3 % less for Cholesterol) compared to the LSTM model, indicating higher prediction accuracy.

Additionally, the hardware prototype facilitates non-invasive measurement and forecasting, offering a promising solution for improved early diagnosis and monitoring. The research highlights the potential of ARIMA and LSTM models in healthcare and diagnostic applications. The superior performance of the ARIMA model in predicting glucose and cholesterol levels positions it as a valuable tool for real-time applications. This study provides a unified solution for non-

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invasive measurement and forecasting, aiming to enhance early diagnosis and monitoring of diabetes and cardiovascular risk.

## Introduction

"Sugar-free & 0 % Trans-fat" are the two buzzwords printed on almost every edible item. This is a clear reflection of the present health scenario. Diabetic Mellitus is a chronic metabolic disorder characterised by increased blood glucose levels, but the primary complication associated with the person with diabetes is its cardiovascular risks. Research reveals that about 75 % of people with diabetes die from CVD caused by blocked arteries (caused by waxy cholesterol deposits). Thus, a diagnosis of blood glucose along with cholesterol levels is meaningful. It is a known fact that the world diabetic population is increasing at a sky-rocketing speed, unknown is that about 50 % of the world population is unaware of their diabetic condition (as per WHO). Painless and Non-invasive diagnostic methods facilitate diagnosis readiness among the population and thus serve as an indirect means of increasing early disease diagnosis. On the other side, considering a diabetic patient, (i) forecasting prevents vulnerability, and (ii) real-time continuous/periodic glucose monitoring satisfies the demand of a Critical Care Unit. These requirements for diagnosis and monitoring of diabetes are the primary motivation behind this research work. The paper is organised with a detailed literature survey in Section II, followed by Section III's methodology and Section IV's experimental results. Section V presents the conclusion.

## literature survey

Diabetes, a metabolic disorder characterized by high blood glucose levels due to the abnormal amount of insulin in the body and Cholesterol, waxy deposits in blood vessels which block the blood flow, are the two primary reasons behind Cardiac arrest in patients. This reveals the importance of maintaining these levels in healthy individuals and their patient monitoring. Thus, blood glucose and cholesterol prediction are important for diabetes treatment and cardiovascular risk prevention. Much research and development in measuring and monitoring glucose and cholesterol levels is being carried out, and it is essential to increase the precision of their measurements further. Monitoring glucose levels in the blood can help detect glucose metabolism abnormalities, improve the effectiveness of related treatments, and prevent severe degenerative consequences from long-term development [1]. According to survey findings, 693 million individuals worldwide are predicted to have diabetes by 2045 [2]. Major causes for cardiovascular disorders – a chief contributor towards world mortality rate- are abnormal cholesterol and glucose levels. Earlier forecast is considered one of the best cholesterol-lowering approaches [3]. Titration of the administered insulin doses is crucial in managing diabetes patients and chronic health conditions, such as Type 1 diabetes, are managed [4], because glucose control and insulin are essential to treating the disease [5]. Continuous Glucose Monitoring (CGM) is an effective strategy for constantly monitoring a diabetic patient's blood glucose level. This method significantly increases the data available to prediction models for forecasting future blood glucose (BG) trends and efficiently negates abnormal events [6,7,8].. Unreliable BG prediction would result from inaccurate CGM values [9] and cause numerous complications [10].

Effectively managing hyperglycemia or hypoglycemia will assist with accurate blood glucose level predictions. [11]. However, various factors may affect glucose variability and cause individuals to respond differently under the same circumstances [12]. Machine learning algorithms are implemented to predict the actual value of BG levels [13], and it is a prerequisite in the current world [14], which will enable quick implementation of preventive steps against health risks caused by abnormal changes. Predictive analytics is an artificial intelligence algorithm that aims to diagnose diseases as accurately as possible, improving patient care, resource optimization, and clinical outcomes [15]. An algorithm trained using actual data collected for 30 days with an interval of 5 minutes is used for glucose series prediction [16]. However, because blood glucose series typically have linear and nonlinear properties, prediction models cannot accurately predict future BG outcomes [17]. Consequently, developing the blood glucose series forecasting model and correctly predicting future time-varying trends has become one of the most demanding applications.

Two models are especially popular in time series analysis and prediction: The AR Integrated Moving Average (ARIMA) model and the Long Short Term Memory model (LSTM) [18]. To predict BG changes during the process of health monitoring and assessment, current prediction models, such as ARIMA and LSTM, are frequently used [19]. An ARIMA model with adaptive orders is set up by Jun Yang et al. to forecast the varied BG concentrations [20] constantly. ARIMA models are one of the most widely used techniques for predicting infectious diseases, including haemorrhagic fever [21], Coronavirus disease 2019 [22], brucellosis, hepatitis, syphilis, influenza, tuberculosis, HIV, as well as blood glucose levels and hypoglycaemia, [23] hospital daily outpatient visits [24], and so on. The use of LSTM, a specific case of recurrent neural networks (RNN), has grown recently in fields like speech recognition, disease prediction, and traffic flow prediction [25,26,27]. LSTM algorithm was effectively used in the quality monitoring and maintenance of natural water bodies [28,29,30]. The LSTM model, in combination with the Gated Recurrent Unit, has been effectively used in predicting river run-off [28]. Water quality prediction by optimized LSTM [29] resulted in good results compared to the ensemble model. Precipitation forecasting [30] is done effectively using an integrated LSTM model. LSTM is mainly used to predict blood glucose concentration [31], and without altering the architecture, LSTM can easily integrate additional features like skin conductance, heart rate, and skin temperature [32]. For time series prediction and forecasting, ARIMA and LSTM are both efficient techniques. By upgrading model parameters, the ARIMA model is created with an adaptive identification algorithm to increase the precision and enforceability of predicting Blood Glucose by adaptation of the time-varying non-stationary CGM data [33]. A hardware model of non-invasive glucose and cholesterol prediction is presented in [34]. An experimental study of the prediction accuracy of different

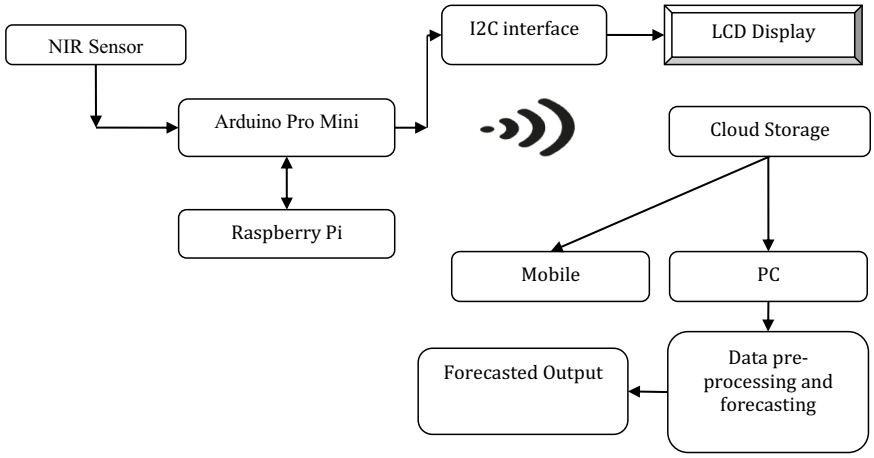



**Fig. 1.** Block diagram representation of the hardware prototype.

algorithms [35], considering a common dataset, revealed that linear algorithms perform well in glucose predictions compared to nonlinear models. The prediction of glucose levels to deliver the required insulin was studied [36,37],. Cholesterol data collected from different patient groups were used for a clustering analysis based on patient grouping [38]. Experiments were conducted to predict Glucose levels based on the nutritional intake of patients [39]. Non-invasive cholesterol detection from saliva [40] and skin [41] is also attempted utilizing nano-bio-sensors. Bio-sensors in the form of contact lenses also serve as a new attempt towards measuring cholesterol non-invasively. LSTM model has introduced RNN and LSTM models for image capturing applications too [42].

Based on the literature reviewed, we could identify certain research gaps with the existing works addressed in our work. The difficulties identified with the present research include (i) Prediction accuracies are limited by inaccurate CGM data [9]. Our hardware prototype is trained based on the correlation functions between the concentration of glucose, lipid-protein and the voltage generated by the sensor for different In-vitro test solutions. A machine learning algorithm determines the correlation. This facilitates a reliable, non-invasive measurement. (ii) Due to the nonlinearity of real BG data [17], it could not predict future BG values. Also, research on developing a unified solution of combined non-invasive measurement and forecasting facility in a single system is very few, and we tried attempting the rare one. The significance and accuracies of ARIMA and LSTM in time series forecasting have motivated us to compare the two models in glucose and cholesterol trend prediction.

This article brings a fresh perspective to the demanding health challenges of diabetes and cardiovascular diseases by introducing a novel, non-invasive monitoring system combining advanced technology and predictive modelling. By developing a hardware prototype that uses Near Infra Red (NIR) sensors for painless measurement of glucose and cholesterol levels, the research departs from traditional invasive assay, making monitoring more patient-friendly. Furthermore, integrating Artificial Intelligence (AI) and Cloud computing allows real-time forecasting of these vital health markers. This approach significantly enhances the proactive potential for healthcare solutions. Moreover, the research conducts a comparative analysis of prediction models, highlighting the superior performance of the Auto-Regressive Integrated Moving Average (ARIMA) model over Long Short-Term Memory (LSTM) in predicting glucose and cholesterol levels. The synthesis of non-invasive measurement and forecasting into a unified system offers a comprehensive solution, addressing the critical needs of early diagnosis and continuous monitoring. This innovative approach has the potential to revolutionize patient care, making it safer and more effective, thereby setting a promising direction for the future of healthcare.

## Methodology

### *Hardware development of non-invasive detecting*

A hardware prototype is developed to measure blood glucose and cholesterol non-invasively from patients. A hardware prototype has been developed for measuring glucose and cholesterol values non-invasively. The measuring unit uses a NIR sensor to make non-invasive sensing possible. Studies have proven that Near Infra Red sensors have distinct absorption patterns for different blood constituents. Thus, an NIR light is made to pass through the patient's finger, and a photodiode is used for detecting the variation in light intensity caused by absorption by cholesterol particles and glucose in the blood. The photodetector output is multiplexed and sent to Raspberry Pi for pre-processing. Correlation between the absorption pattern and actual In-vitro measurement is used for training, and thus, the non-invasive determination of glucose and cholesterol is feasible with good accuracy.

The hardware unit thus designed is used for determining the glucose and cholesterol data of a patient under monitoring for a period of one month, and the values are updated for predicting the time series forecasting of glucose and cholesterol variations in the patient. Then, the stored glucose and cholesterol values are formed as a glucose and cholesterol dataset, which has the time and measured values in separate columns. Then, the data is pre-processed and divided into a training set and test set using machine learning

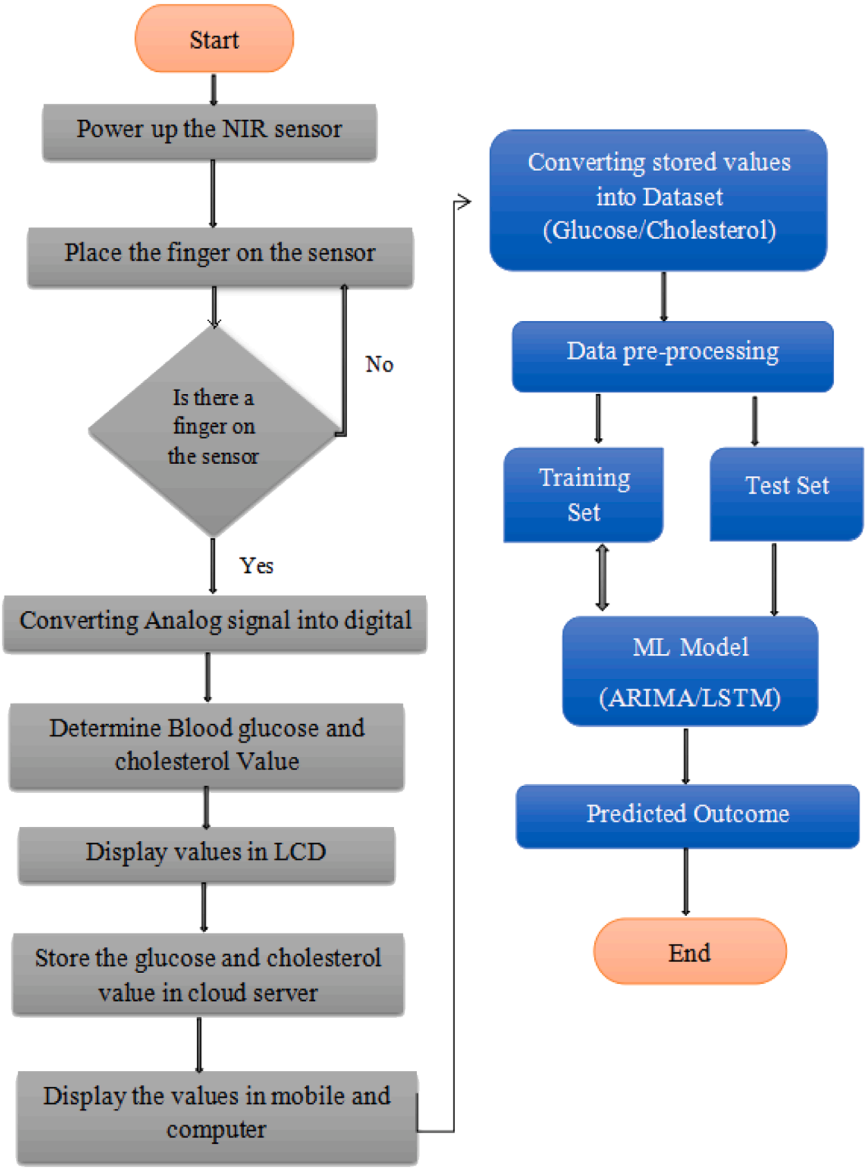



**Fig. 2.** Flow chart depicting the workflow. The Gray coloured portion of the flowchart is referenced from [34] to depict the overall workflow (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

algorithms autoregressive integrated moving average (ARIMA) model and long short-term memory networks (LSTM) model the glucose and cholesterol values are predicted. Fig. 1 shows the block diagram representation of the hardware prototype used for measuring glucose and cholesterol non-invasively, and Fig. 2 depicts the flowchart for the process.

*Workflow*

The working methodology of the proposed system is represented in Fig. 2. After absorption is detected using a photodiode, the NIR light and the received analog signal are converted to a digital signal. A correlation mapping of glucose and cholesterol data with the absorption spectrum aids the non-invasive measurement of glucose and cholesterol. In addition to displaying the measured values, they are stored in a cloud server, which can be accessed anytime by mobile or computer. The values monitored for over a month are stored in the cloud. This data is converted to a time series format for training the AR Integrated Moving Average model and Long Short Term Memory models. The real-time data obtained from our hardware prototype are represented in Figs. 3 and 4. Data in time series format is represented by Figs. 5 and 6. The input data set is divided into training and testing datasets in a 70 % and 30 % ratio.




**Fig. 3.** Blood glucose data stored in cloud.

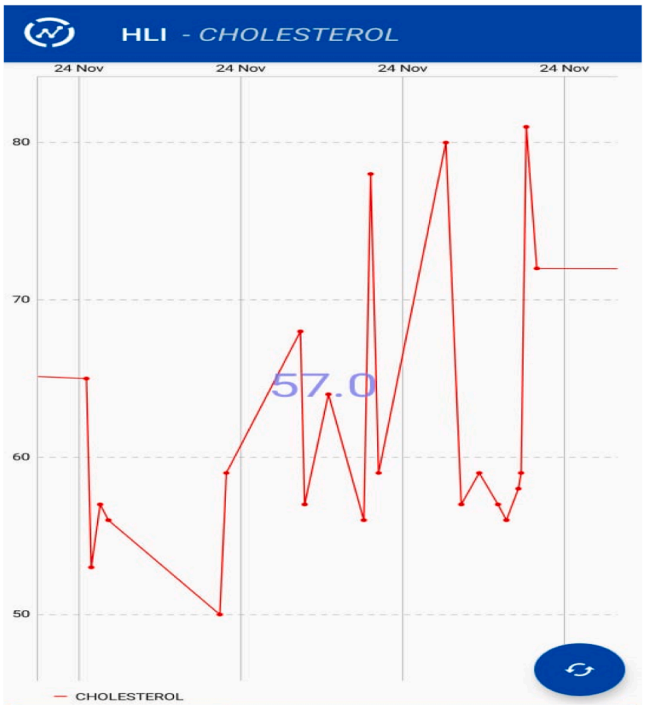



**Fig. 4.** Blood cholesterol data stored in cloud.

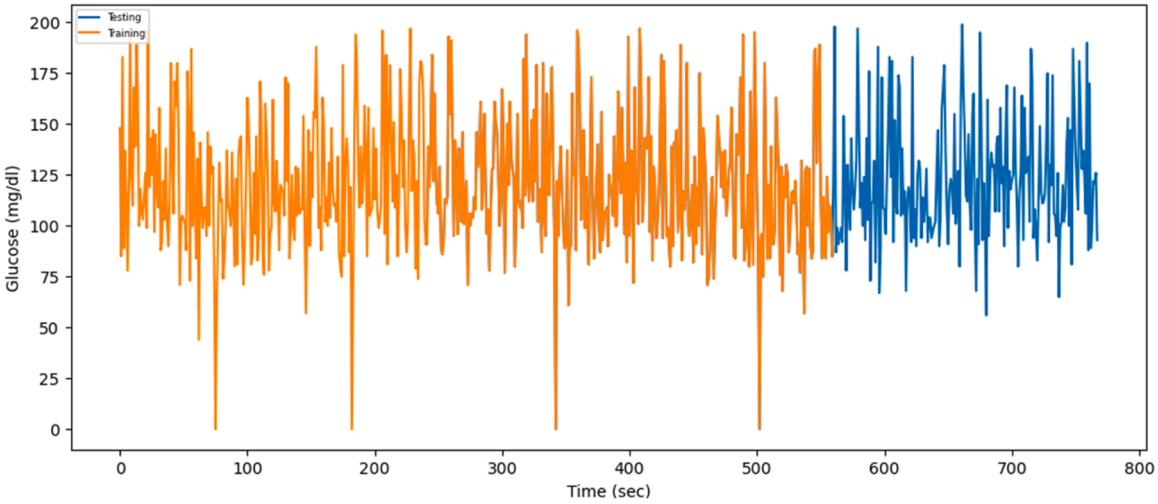



**Fig. 5.** Blood glucose data.

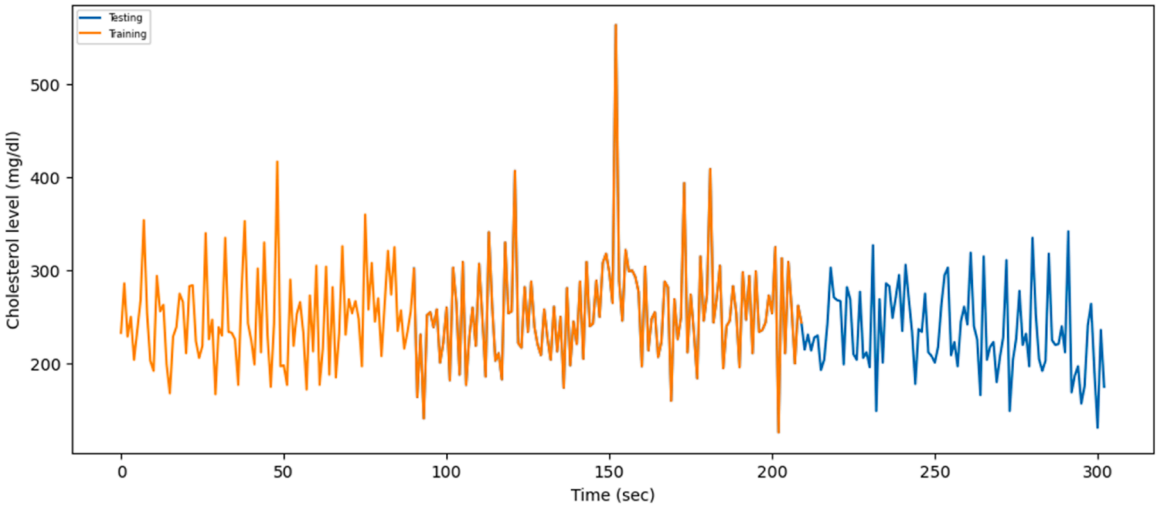



**Fig. 6.** Blood cholesterol data.

## Experiment and results

### *The experimental data*

To predict the glucose and cholesterol levels for a patient, sample data are required. These are formed by a patient's glucose and cholesterol levels measured for a month, which are stored in a cloud server, as shown in Figs. 3 and 4.

In this experiment, to assure the reliability of the test data, blood lipid and glucose data monitored for one month and stored in cloud by the non-invasive measurement kit, which constitutes numerous values of glucose, cholesterol and time are used. As shown in Figs. 5 and 6, the graph has been drawn with the created dataset with respect to glucose, cholesterol values and time. The x-axis represents the time, and the y-axis represents the glucose and cholesterol values in Figs. 5 and 6.

### *Autoregressive integrated moving average (ARIMA) model*

Box and Jenkins introduced the Autoregressive Integrated Moving Average process model (ARIMA), a linear time series analysis and forecasting model. Auto-Regressive (AR), Moving Average (MA), Seasonal Autoregressive Integrated Moving Average (SARIMA), and other models are all included in the ARIMA model. Three parameters must be specified for time series analysis they are moving average order (q), difference order (d), and autoregressive order (p). This study uses this prediction-based algorithm to forecast cholesterol and blood glucose levels. The Autoregressive Integrated Moving Average (ARIMA) model is created with an adaptive

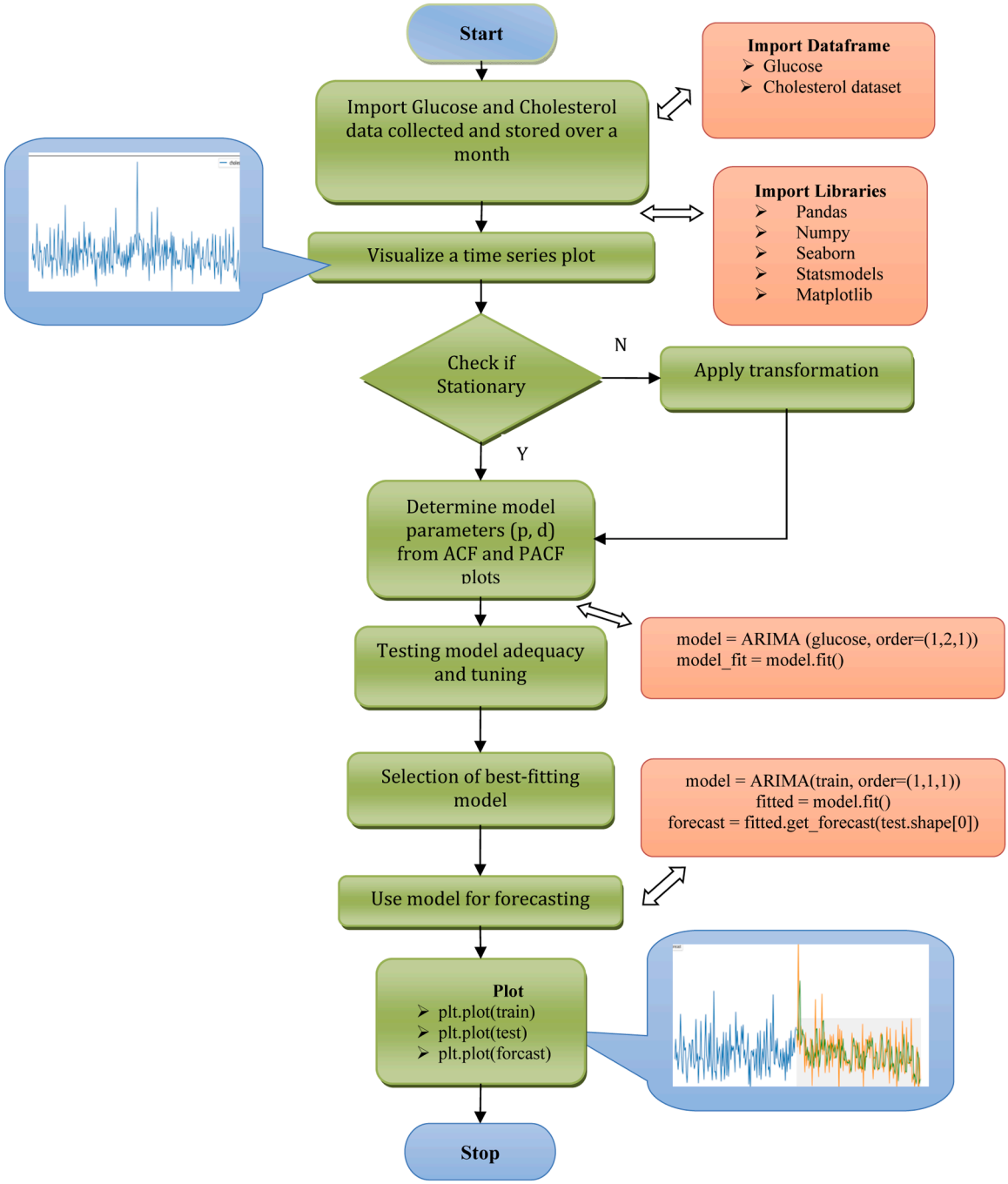



**Fig. 7.** Flow diagram of ARIMA Algorithm.

identification algorithm to increase the precision and enforceability of predicting blood glucose and cholesterol by adopting the time-varying non-stationary continuous glucose and cholesterol data. Flow diagram of ARIMA algorithm shown in Fig. 7.

The Implementation of the ARIMA prediction algorithm starts with importing libraries such as pandas, numpy, seaborn, statsmodel and matplotlib. Then, import the prepared dataset as a csv file. ARIMA is imported from statsmodels, and the dataset is segregated into two parts: one is a training set, and another one is a testing set. By fitting the ARIMA model, glucose and cholesterol values are predicted. The predicted values are illustrated as a graphical representation using the plot function.

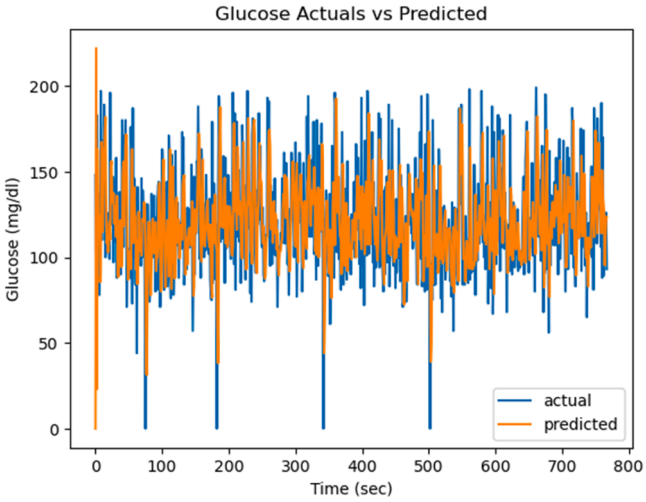



**Fig. 8.** Actual glucose vs predicted glucose.

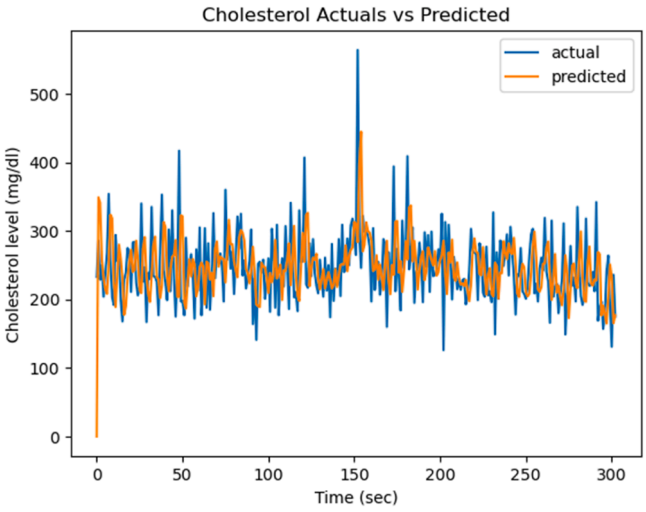



**Fig. 9.** Actual cholesterol vs predicted cholesterol.

*ARIMA model prediction*

Figs. 8 and 9 represents the actual value of glucose cholesterol and the predicted values of glucose and cholesterol in upcoming time with respect to the ARIMA (1,2,1) prediction algorithm. It depicts the actual data (blue line) and predicted data (red line). Based on the predicted values, reliable actions are taken to prevent hyperglycemia or hypercholesteremia.

*ARIMA model forecast*

The significance of the ARIMA model algorithm is its capability of forecasting outputs based on time series data training, and thus, a forecast of the trend in glucose and cholesterol variations in patients can be predicted. Figs. 10 and 11 shows the forecasted glucose and cholesterol predictions represented in green colour. And the orange colour represents the actual Glucose and cholesterol values in Figs. 10 and 11, respectively.

*Long short-term memory networks (LSTM) model*

LSTM, a specific recurrent neural network topology type, maintains short-term memories (RNN). It is frequently applied to challenges dependent on time series data, including speech recognition, handwriting recognition, healthcare pathway prediction, etc. Because it uses the "remember" mechanism through a succession of gates, LSTM is uniquely suited to manage long-term dependencies

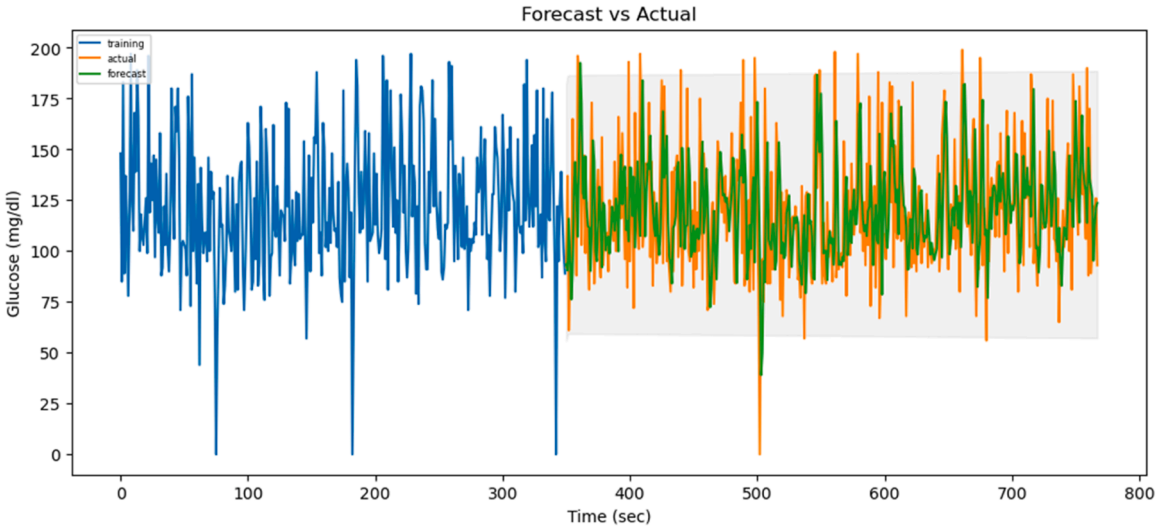



**Fig. 10.** Glucose Forecasting based on the time series data.

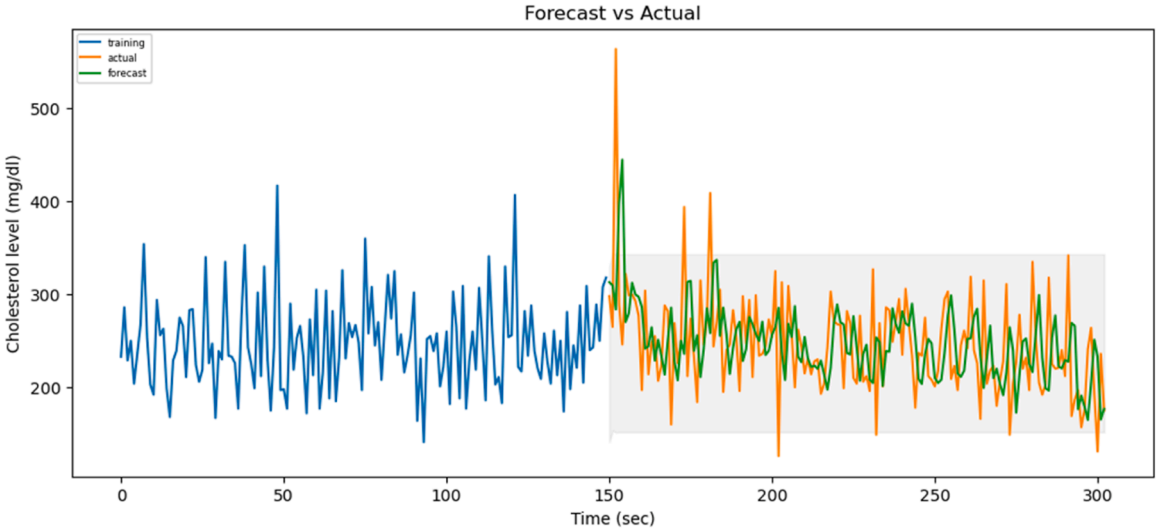



**Fig. 11.** Cholesterol Forecasting based on the time series data.

in contrast to ordinary RNNs. This aspect is appropriate for the glucose prediction problem's possible prolonged observation window, which challenges other machine-learning techniques.

The LSTM prediction algorithm starts with importing libraries such as pandas, numpy, seaborn, statsmodel, math, datetime and matplotlib. Then import the prepared dataset as a csv file. The dataset is segregated into two parts: one is a training set, and another one is a testing set. By fitting the LSTM model, glucose and cholesterol values are predicted. The predicted values are illustrated as a graphical representation using the plot function. LSTM can remember values from previous stages for future use, useful for predicting time sequences. It is mainly used to predict blood glucose concentration, and without altering the architecture, additional features such as skin conductance, heart rate, and skin temperature can be added to the LSTM model. Flow diagram of LSTM shown in Fig. 12.

As the number of epochs increases sequentially, the LSTM's output becomes more accurate. Figs. 13 and 14 represents the epoch vs loss plot of the LSTM model.

*LSTM model prediction*

Figs. 15 and 16 compare the actual values (blue) of glucose and cholesterol to the predicted values (red) of glucose and cholesterol in future obtained from the trained LSTM Neural Network model.

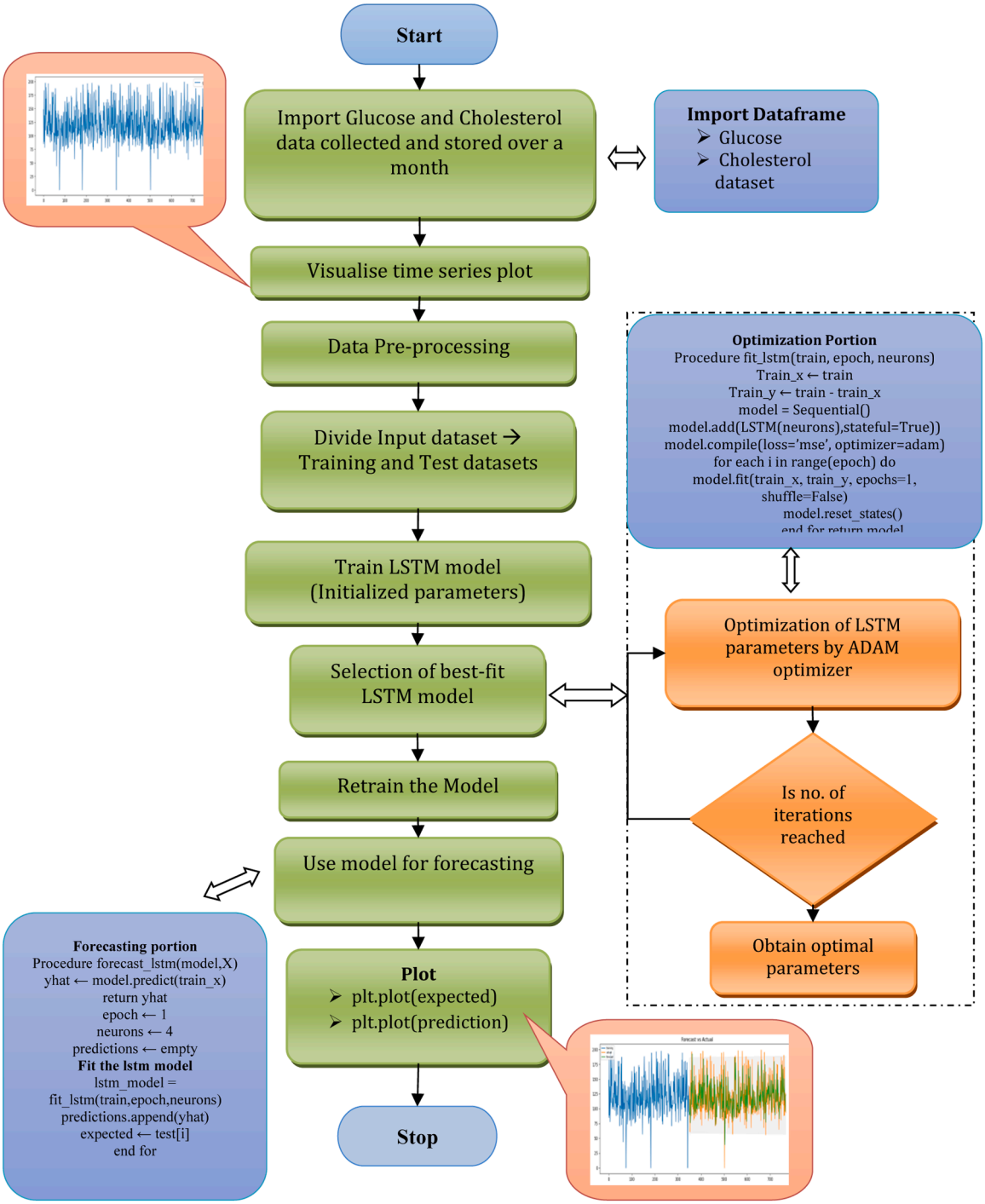



**Fig. 12.** Flow diagram of LSTM Algorithm.

*Comparison between ARIMA and LSTM models*

Figs. 17 and 18 represent the actual values and predicted values with respect to prediction-based algorithms ARIMA and LSTM. The continuous blue line denotes the actual data, the dotted green line depicts the ARIMA model-based prediction, and the dotted red line depicts the LSTM model-based prediction. The ARIMA model is better than the LSTM model in prediction, in which the actual value is nearer to the ARIMA model predictions. ARIMA model has greater accuracy in tracking real-time data changes. Also, data forecasting is possible with the ARIMA model.

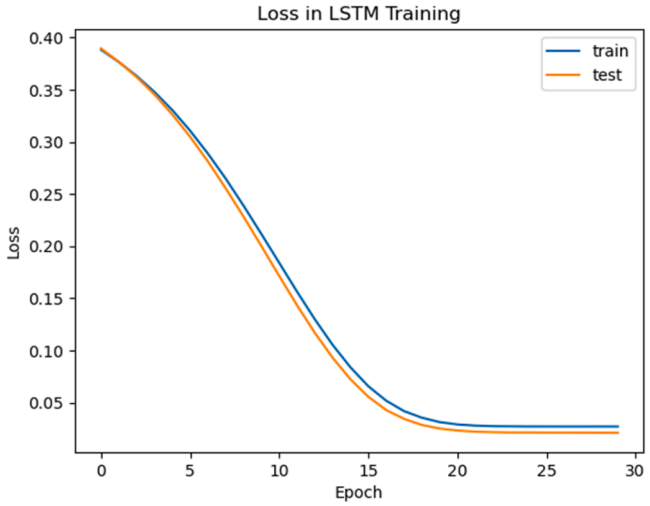



**Fig. 13.** Epoch vs Loss in LSTM for glucose prediction.

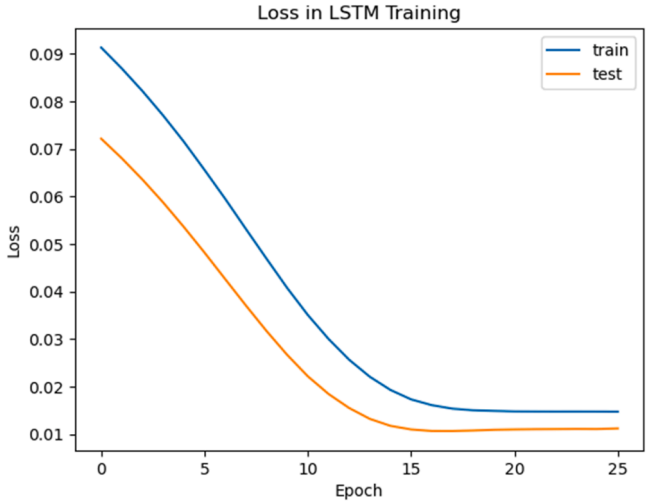



**Fig. 14.** Epoch vs Loss in LSTM for cholesterol prediction.

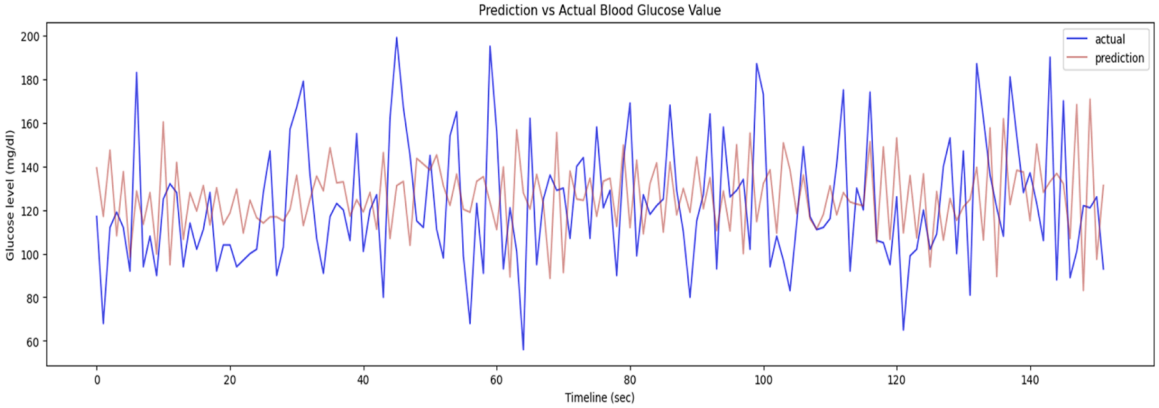



**Fig. 15.** Actual vs prediction glucose in LSTM.

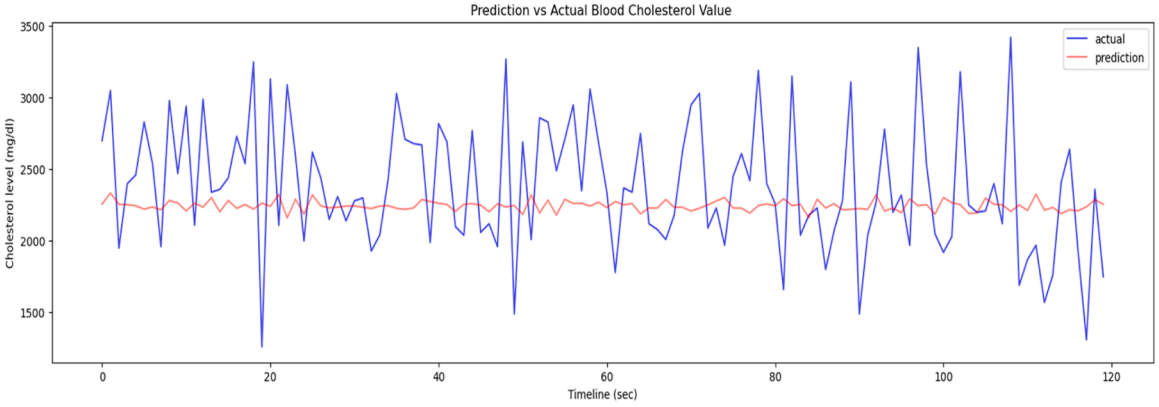



**Fig. 16.** Actual vs prediction cholesterol in LSTM.

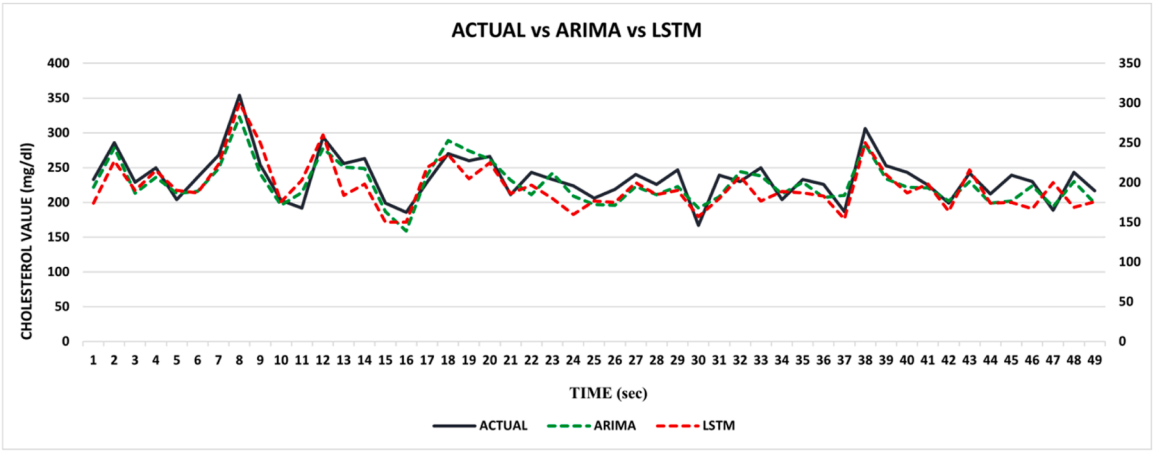



**Fig. 17.** Actual vs Predicted models.

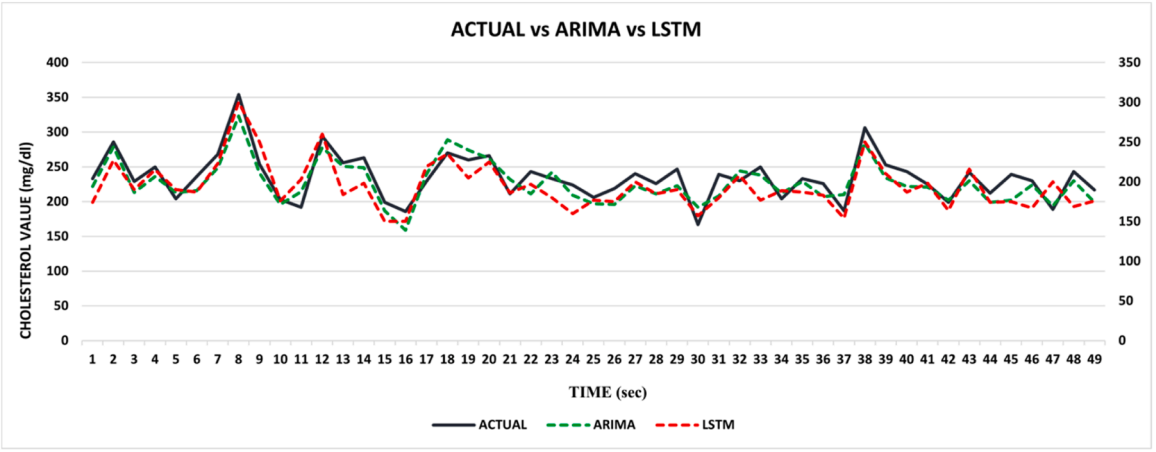



**Fig. 18.** Actual vs Predicted models.

*Root mean square error*

Figs. 19 and 20 represents the graph depicting the error metric as RMSE value between the two models. Root Mean Square Error (RMSE) assesses the prediction performance models. The prediction algorithm exhibits a low or zero root mean square error value and

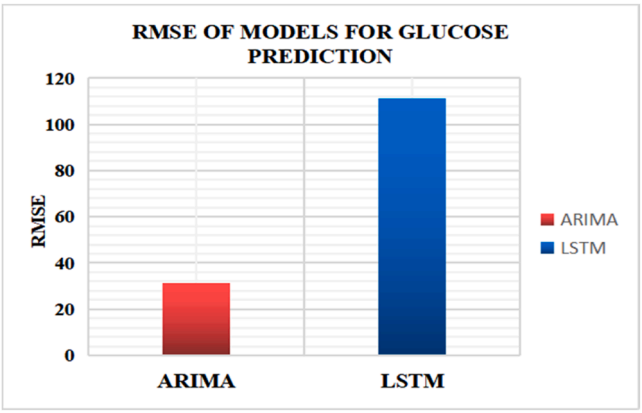



**Fig. 19.** RMSE of models for glucose prediction.

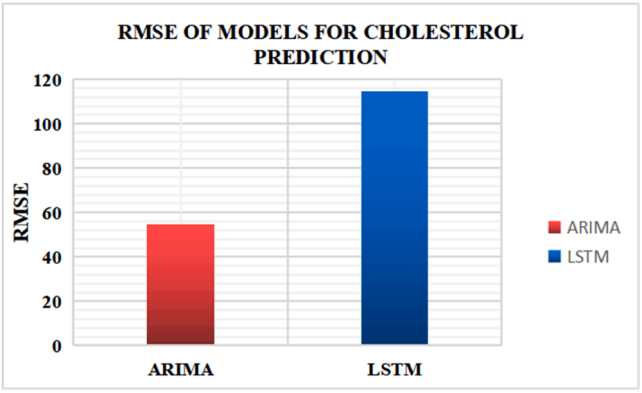



**Fig. 20.** RMSE of models for cholesterol prediction.

**Table 1**
RMSE of ARIMA and LSTM model predictions.

| Model | Predicted value | RMSE |
|---|---|---|
| ARIMA | Glucose | 31.24 |
| | Cholesterol | 54.62 |
| LSTM | Glucose | 109.43 |
| | Cholesterol | 110 |

is considered a better prediction model. Here ARIMA (Autoregressive Moving Average) model exhibits a lower root mean square error value than the LSTM (Long Short Term Memory) model. A comparison of the RMSE values of the ARIMA (Autoregressive Moving Average) model and LSTM (Long Short Term Memory) model is presented in Table 1. Root Mean Square Error measures the forecasting efficiency of the two models used in glucose and cholesterol forecasting. RMSE reveals how close the predicted result is to the observed data points.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(F_i - A_i)^2}{N}} \tag{1}$$

## Conclusion

This paper empirically analyses the blood glucose series of experimental samples. Machine learning-based neural network models have been implemented to forecast blood glucose and cholesterol levels. The prediction-based algorithms ARIMA (AR Integrated Moving Average) and LSTM (Long Short Term Memory) are the models used to predict blood glucose and cholesterol levels at various time-varying data. The online ARIMA model was used to forecast real-time dynamic blood glucose and cholesterol variations. The LSTM can better capture the input sequence's present and past context and learn to outline key features, remove background noise, and

capture more long-term dependencies. There exists a prediction comparison between the ARIMA and LSTM models. Since a real-time experiment is conducted to validate the predictions based on the CGM data, it is inferred from the experimental results that the ARIMA prediction model can successfully improve prediction accuracy than LSTM. The application of machine learning models opens up numerous opportunities for clinicians and helps enhance millions of people's health, safety, and quality of life.

## Declaration of Competing Interest

Current paper has no conflict of interests

## Data availability

Data will be made available on request.

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